

# SUE LAB

## BUILD DOCUMENTATION

### 12U OPEN FRAME SERVER RACK

**Date:** September 2026

**Status:** Completed

#### 01 — OVERVIEW

The VEVOR 12U Open Frame Server Rack is the physical foundation of Sue Lab. It provides a centralized structure for mounting and organizing networking and cybersecurity equipment while keeping components accessible for installation, configuration, and maintenance.

The open-frame design allows easy access to equipment ports and cabling while supporting current equipment and future additions to the lab.

#### 02 — SPECIFICATIONS

- **Manufacturer:** VEVOR
- **Model:** HT-W6412
- **Rack Height:** 12U
- **Adjustable Depth:** 23–40 inches
- **Mounting Width:** Standard 19 inches
- **Ground-Mounted Load Capacity:** 500 lbs
- **Wall-Mounted Load Capacity:** 150 lbs
- **Material:** Carbon steel
- **Color:** Black
- **Net Weight:** 31.2 lbs / 14.15 kg
- **Product Dimensions:** 23 × 20.1 × 24.6 inches
- **Casters:** 4 included
- **Shelves:** 2 included
- **Grounding Wire:** Included

#### 03 — WHY I CHOSE IT

I chose the VEVOR 12U Open Frame Server Rack to provide a dedicated physical structure for Sue Lab. Its 12U capacity provides enough space for networking, compute, firewall, and power equipment while leaving room for future expansion.

The adjustable depth and open-frame design also make it practical for accessing equipment and managing cables during the build. Overall, this server rack gives me everything I need to start my journey while staying budget-friendly.

## 04 — INSTALLATION / CONFIGURATION

The rack was assembled using the included hardware and components. The frame, shelves, and casters were installed to create a mobile, free-standing equipment rack.

After assembly, the rack was positioned as the physical foundation of Sue Lab and prepared for the installation of networking and other lab equipment.















## 05 — FINAL SETUP

The completed 12U rack provides the physical foundation for Sue Lab. The two included shelves provide additional space for equipment that is not directly rack-mounted, while the casters allow the rack to be moved when necessary.

The rack is now ready for the installation of the Cisco Catalyst switch and additional infrastructure as the lab continues to develop.





## 06 — CHALLENGES

- **Challenge:** Installing and securing the shelves during assembly.
- **Solution:** Adjusted the shelf positioning and hardware alignment to ensure the screws properly reached the rack mounting points.
- **Challenge:** Installing and tightening the casters.
- **Solution:** Secured each caster to the bottom of the rack and tightened the hardware to ensure the rack remained stable while allowing mobility.

## 07 — WHAT I LEARNED

- Learned how to assemble and configure a 12U open-frame equipment rack.

- Learned how rack shelves and mounting hardware are positioned within a standard 19-inch rack.
- Gained hands-on experience preparing a physical infrastructure environment for networking equipment.
- Learned the importance of planning equipment placement before installing additional hardware.

## **08 — NEXT STEPS**

The completed rack provides the physical foundation for the next stage of Sue Lab. The next steps are to install the Cisco Catalyst switch and continue adding networking, firewall, compute, and power equipment as the lab develops.